

Challenges (4): modeling frameworks

Some platforms exist(ed...)



COVID-19 Pharmacology
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Viral Kinetic Model

Viral dynamics models provide valuable insight into the life cycle of infectious agents. This in silico viral kinetic model reveals the challenges of monotherapy treatment: a monotherapy treatment must be initiated very early in viral infection and be quite potent. However, combination therapy where different steps of the SARS-CoV-2 life cycle are modestly impacted may be an alternate strategy. Numerous model structures have been proposed, but the model for extremely fast and short-duration replication (e.g., influenza) derive from Baccam 2006. The model used here is from Gonçalves 2020 and expands the infected cell population.

We performed simulations with this viral kinetic model and found that combinations of therapeutics targeting specific rate constants have greater probability of efficacy, especially in the treatment of early infection phase in COVID-19 patients. By targeting multiple points central to viral replication within infected host cells or release from those cells, we may be able to reduce both viral load and host cell infection. Access the full manuscript which is available as a preprint: Michael Dodds, Rajesh Krishna, Antonio Goncalves, Craig Rayner. Model-Informed Drug Repurposing: Viral Kinetic Modeling to Prioritize Rational Drug Combinations for COVID-19. Authorea. June 12, 2020. DOI: 10.22541/au.159200535.51965254Hi

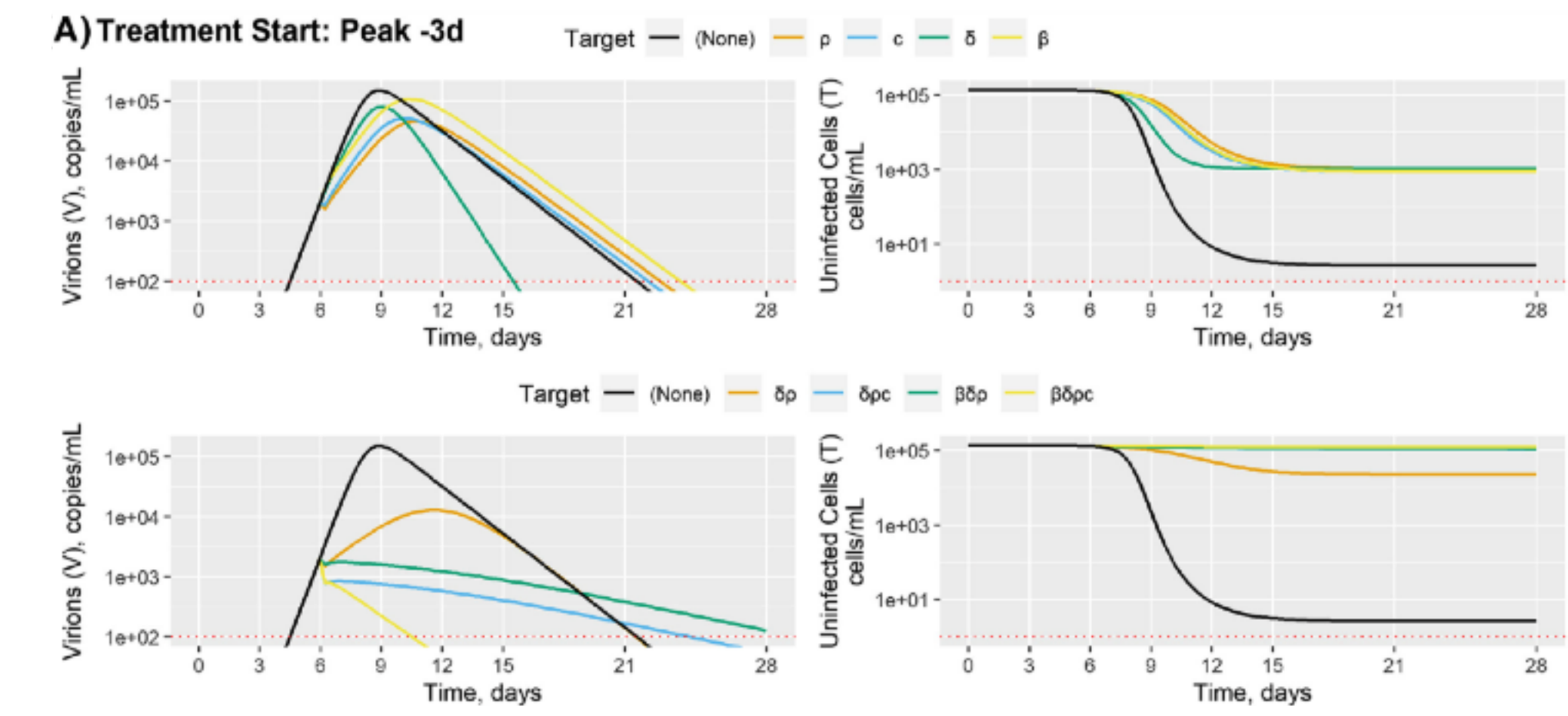


BJCP British Journal of
Clinical Pharmacology

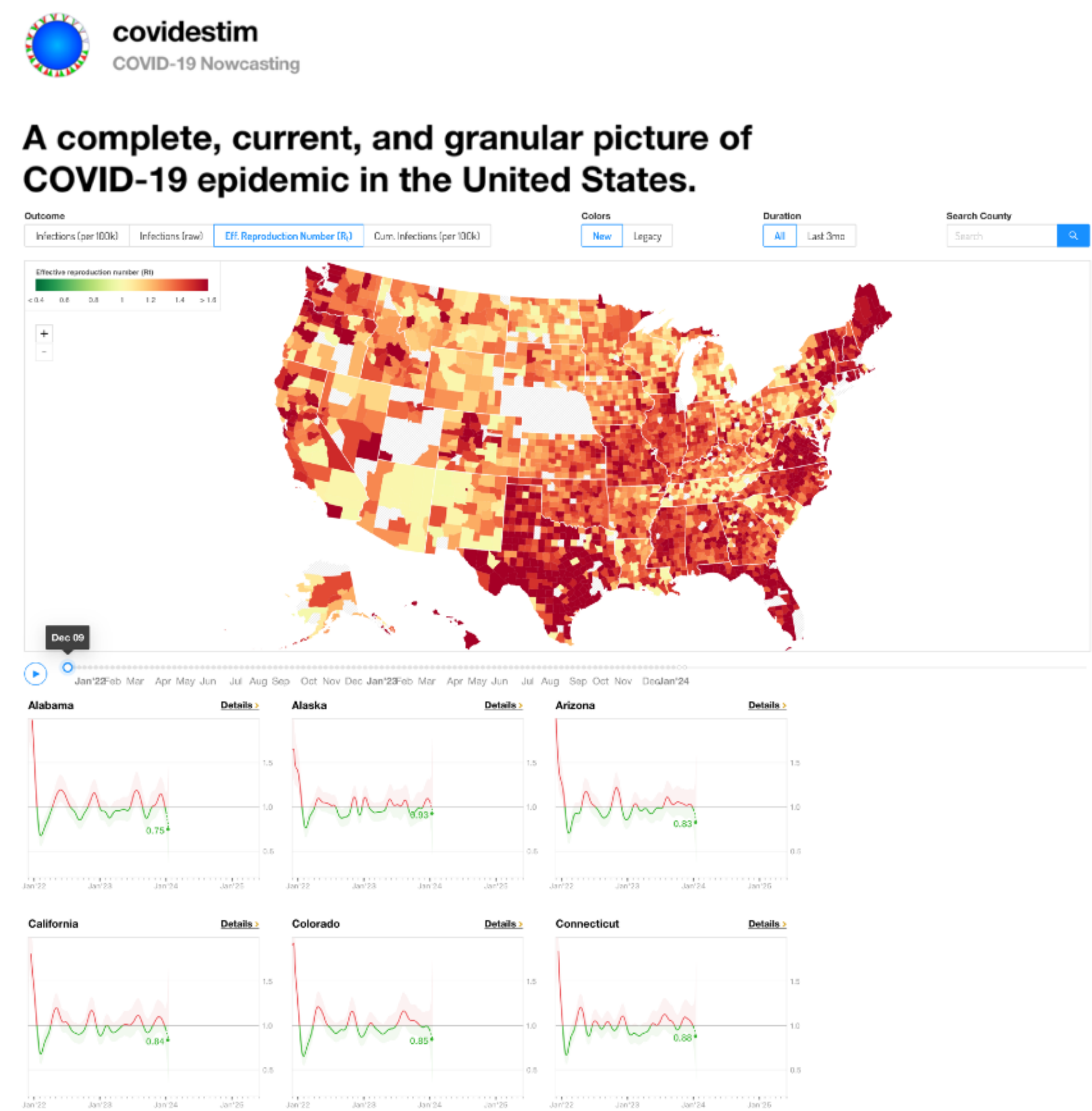
Model-informed drug repurposing: Viral kinetic modelling to prioritize rational drug combinations for COVID-19

Michael G. Dodds, Rajesh Krishna, Antonio Goncalves, Craig R. Rayner

First published: 21 July 2020 | <https://doi.org/10.1111/bcp.14486> | Citations: 22

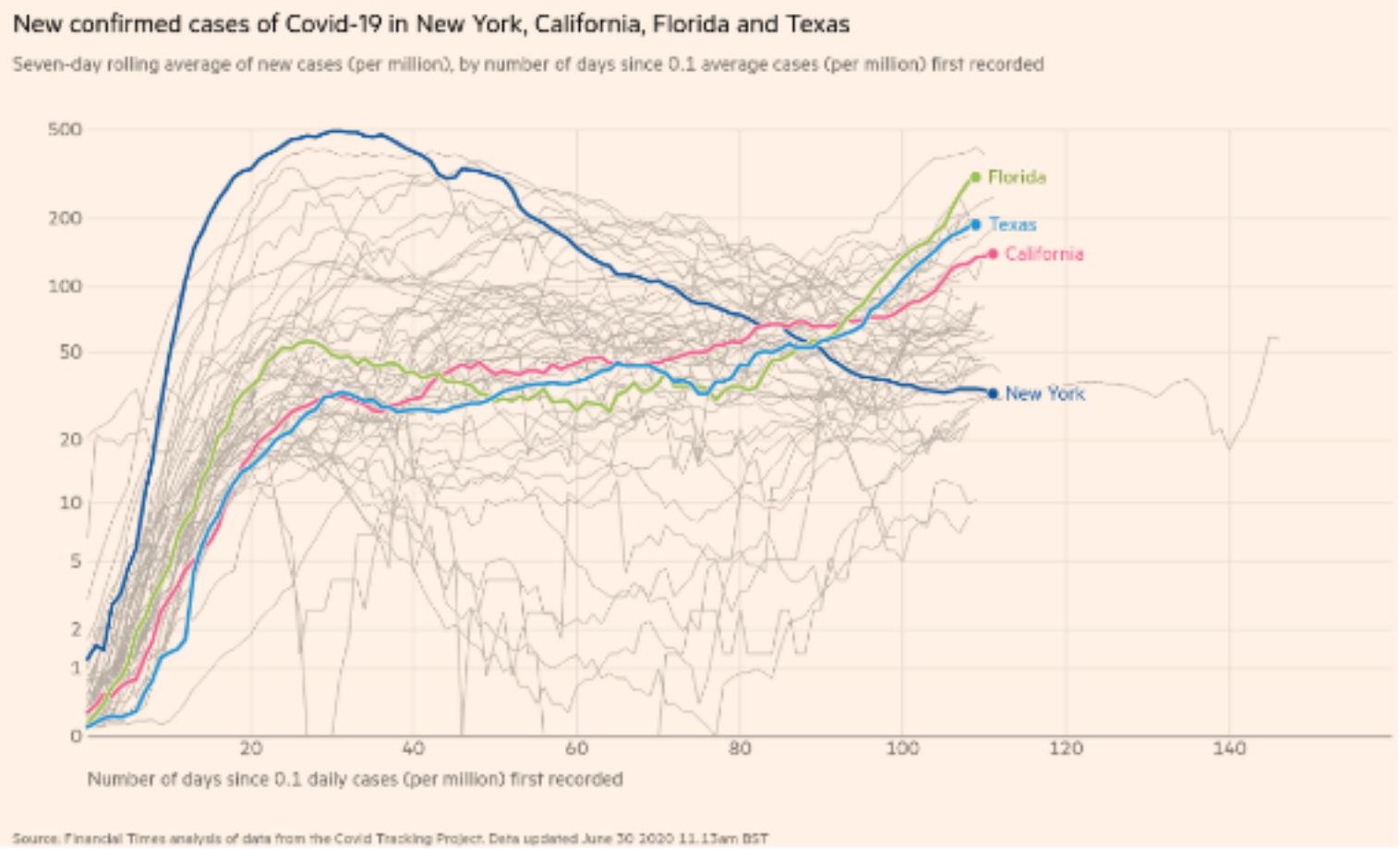
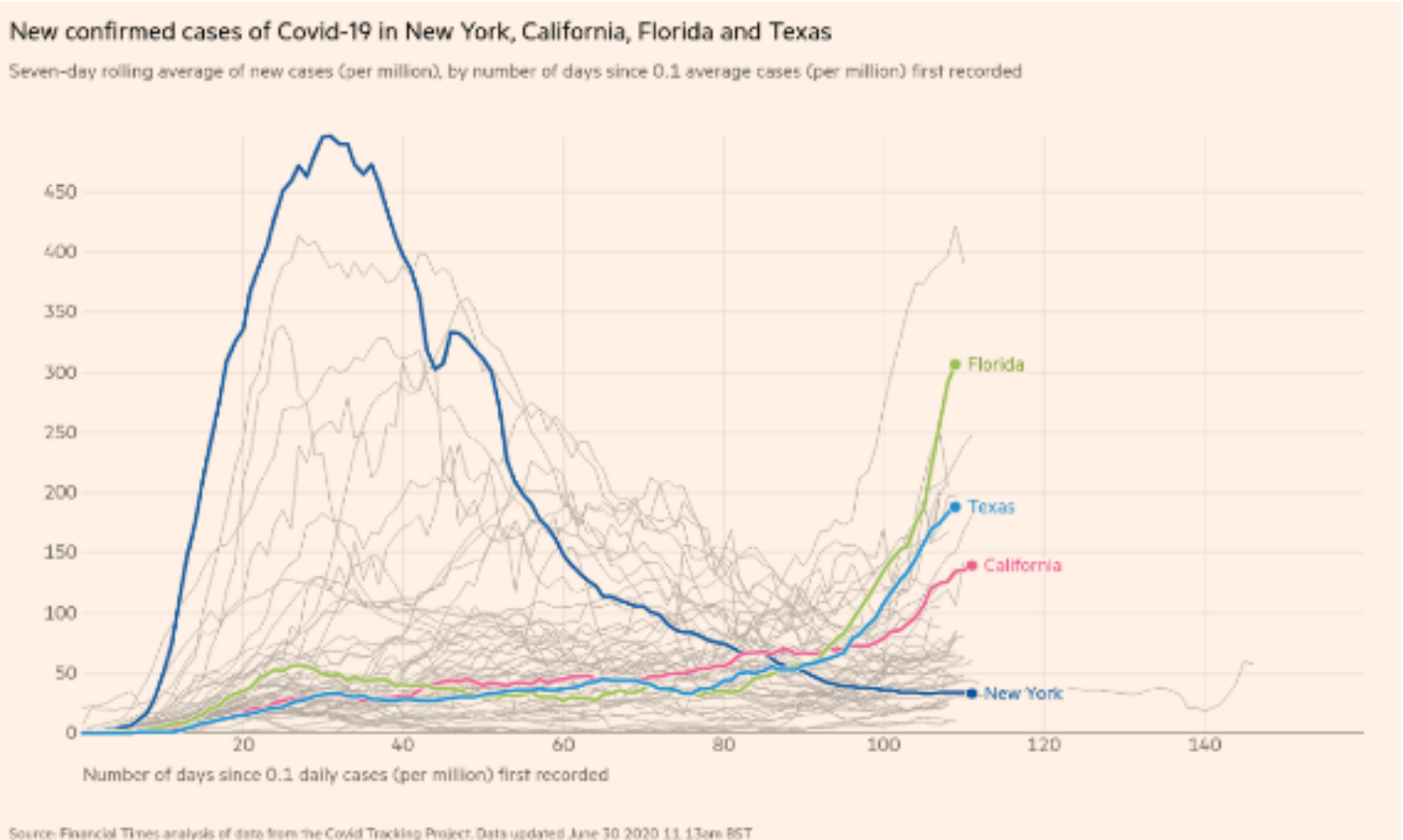


Rt estimation a useful paradigm



Challenges (5): reporting and visualization

Scale?



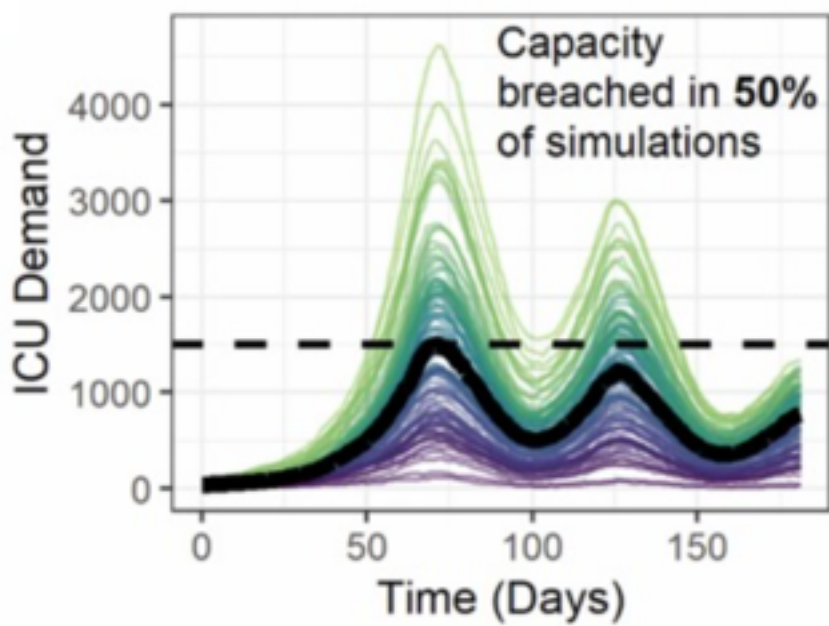
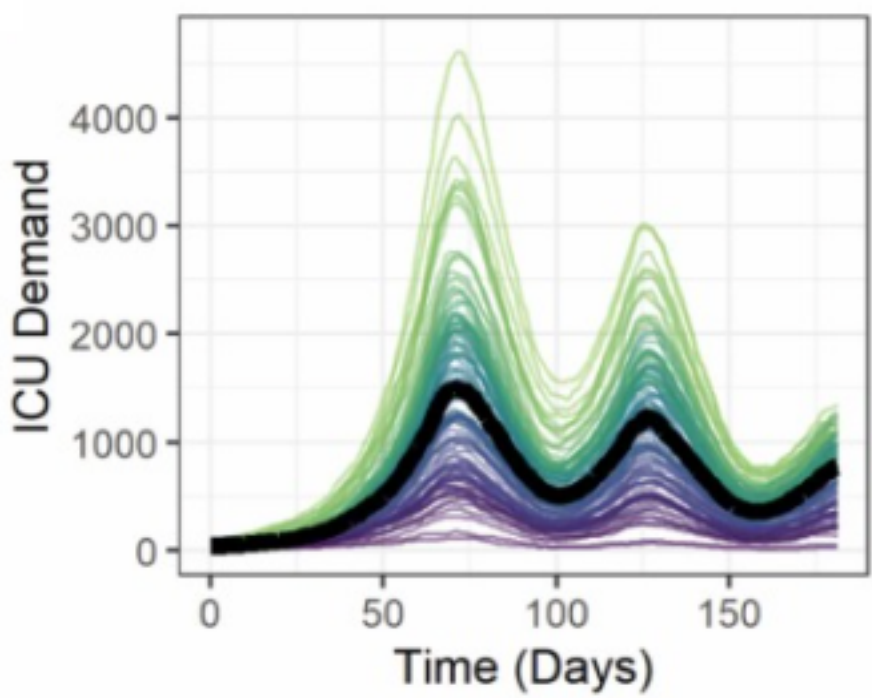
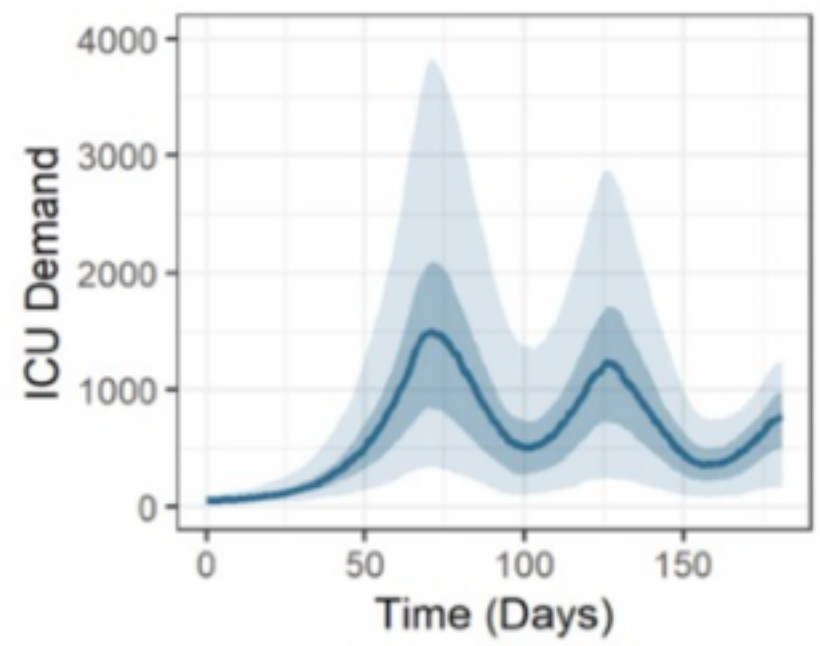
Uncertainty/impact?

Epidemics

Volume 37, December 2021, 100520

Communicating uncertainty in epidemic models

Ruth McCabe ^{a b c}  , Mara D. Kont ^c, Nora Schmit ^c, Charles Whittaker ^c, Alessandra Løchen ^c, Patrick G.T. Walker ^c, Azra C. Ghani ^c, Neil M. Ferguson ^{c d}, Peter J. White ^{c d e}, Christl A. Donnelly ^{a b c d}, Oliver J. Watson ^c



Comparisons?

nature communications

Viral kinetics of sequential SARS-CoV-2 infections

Stephen M. Kissler, James A. Hay, Joseph R. Fauver, Christina Mack, Caroline G. Tai, Deverick J. Anderson, David D. Ho, Nathan D. Grubaugh & Yonatan H. Grad 

Nature Communications 14, Article number: 6206 (2023) | Cite this article

